

Mandara Bhargava

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ACADEMIC QUALIFICATION

Master of Data Science, The University of Queensland, Queensland, Australia (Feb'26 - Present); GPA: 6.5/7

Bachelor of Technology in Electronics and Communication (major) and **Computer Science** (minor), People's Education Society University (PESU), Bangalore, PES University; CGPA: 7.44/10; Aug'25

WORK EXPERIENCE

AI Developer, Emids Technology, Bangalore

Jan'25–Jan'26

(Emids Technology is a US Healthcare company focused on delivering competitive services to healthcare and life sciences in the domains of AI, Data Analytics, and Data Engineering.)

- Functional lead and Client-facing for Software Engineering in the AI team with agentic projects like User Story Gap Analyzer, Developer Code Reviewer to check for compliancy, consistency, and more.
- Led a team of 3 interns to fine-tune a Large Language Model for an automated Mental Health Hotline Operator, reducing manual evaluation time from 45-60 minutes to under 5 minutes using Retrieval Augmented Generation and a knowledge database.
- Developed a Data Mapper Copilot to accurately map healthcare data across domains (Claims, Providers, Administrative) with 98% accuracy, leveraging Retrieval-Augmented Generation, Agentic Workflow, and LiteLLM.

Full Stack Web Developer, 0101 Digit All Solutions, Bangalore

Jun'24–Dec'24

(0101 Digit All Solutions is an AI-oriented company that aims to automate business solutions with Artificial Intelligence to enhance operational and customer experience of clients.)

- Designed the signature product “#Job”, a website built with MERN Stack, to automate the process of finding the right job without the hassle of a resume and prolonged hours of job-search
- Developed a platform, PromoJin, with MERN Stack to host influencer and business campaigns to engage more market interaction and help in an overall increase in revenue generation for all the parties involved, alike

ACADEMIC/STARTUP PROJECTS

Title: Detection of Diabetic Nephropathy using Generative Adversarial Network (GAN) and Artificial Intelligence

Duration: Feb'25 - August'25

Team size: 2

Summary: By detecting and analysing the probability of having Diabetic Nephropathy (DN) in diabetic patients, the project aims to utilise the capabilities of Artificial Intelligence (AI) by generating synthetic reports and imaging using Generative Adversarial Network (GAN) after training on thousands of kidney Ultrasounds, Doppler Renal Scans, MRI and CT scans.

Individual Contribution: Developed a GAN network and trained it to generate relevant datasets after thorough research in the domain. Researching and co-creating Agents and Deep Learning models to work with a prediction of Diabetic Nephropathy

Title: Reliable V2V Computation Offloading Using Deep Reinforcement Learning

Duration: Sep'23 – Dec'24

Team size: 4

Summary: Computational Offloading between vehicles is an emerging paradigm where vehicles deal with resource-intensive and latency-sensitive applications. The paper's approach proposed a Deep Reinforcement Learning (DRL) model based on Deep Deterministic Policy Gradient (DDPG) algorithm. The problem was considered as a Markov Decision Process (MDP) with the Road Side Unit (RSU) / Base Station (BS) acting as a proxy client model trained to perform as server. The model outperformed tested models like the Soft-Actor-Critic (SAC) by 7%, on average with 80% success rate for task completion within least tolerable task delay, in a secure and reliable environment, especially in low-density traffic.

Individual Contribution: An extensive literature survey resulted in quality path loss modelling. Planned the vehicular environment and structured the logic of the algorithm. Additionally, compared the paper's proposed method against a tested Soft Actor-Critic model.

Title: UnJam - Startup

Duration: Nov'19 - May'21

Team size: 1

Summary: A startup proposal for immediate mechanical and emergency response towards broken vehicles to avoid traffic congestion. An integrated SOS for mechanical help, towing services for breakdowns, along with police intervention during accidents. The insurance checking policy was an additional feature added to attract police cooperation. The project placed top 7 out of 124 teams from college and top 20 within colleges from India in Feb'23.

PUBLICATIONS

Best Paper Award, *Mandara Bhargava*, “Reliable V2V Computation Offloading Using Deep Reinforcement Learning” soon-to-be-published in International Conference on Man-Machine Systems (ICoMMS) – Internet of Everything, Aug'25.

AWARDS & ACHIEVEMENTS

- United Nations Sustainability Development Goals Micro-Credential Certification - Mar'26
- Cisco ThingQBator Cohort 5 – Top 7 of 124 teams from PESU, Jul'22
- DAC (Distinction) Scholarship with 7.95/10 SGPA in B. Tech for Semester 1, Jul'22
- College Ambassador for PES University in Cisco ThingQBator for getting the highest number of student registrations, 324 students, in Cohort 5, May'22
- Top 10 teams in Gusty Entrepreneurship 3.0, startup idea competition for in-college students, presented an idea for RFID installed library books for human-free book borrowing, Oct'21
- Class 10th School Topper with 96.2% for Top 100 schools in Karnataka, May'19

WORKSHOPS

- Participated in an online workshop on “Git and Github Tools” conducted by PES University in collaboration with GeeksforGeeks at PES University, Bangalore, Jun'22
- A three-day workshop on “Cloud Computing Bootcamp with Google Cloud Platform” conducted by PES University I/O, Bangalore, Oct'21

PRESENTATIONS

- Presented a 5-day seminar for the Centre for Innovation and Entrepreneurship (CIE) to roughly 1,800 freshman Engineering students in the Bootstrap program at PES University, Bangalore, Aug'23
- Presented a 5-day seminar on Wireless Network Communications (WiNC) in the department of Electronics and Communications to roughly 1,800 freshman Engineering students in the Bootstrap program at PES University, Bangalore, Aug'23 and Sep'22

SEMINARS & CONFERENCES

- Attended a seminar on “Technical Symposium” and “Wireless Network & Cloud Security” by Nokia Bangalore University Collaboration at Nokia, Bangalore, Aug'23
- Participated in a seminar on “Digital Transformation - Table stakes?!” by Raghu Chinnalli, PES University, Bangalore, India, Feb'22

TECHNICAL SKILLS

- Languages: Python, C, Javascript
- Web Technologies: HTML5, CSS, TypeScript, Bootstrap, Express.js, Node.js, React.js, Streamlit
- Database: MySQL, PostgreSQL, MongoDB
- AI/ML Stack: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, SciPy, Hugging Face
- GenAI & RAG: LangChain, Google ADK, Prompt Engineering, RAG Pipelines, Multi-Agent Workflows, LLM Fine-tuning

EXTRACURRICULAR ACTIVITIES

- Chosen as a startup mentor for 5 teams participating in CIE Ignite Season 2, March'26 – May'26
- Held creative journalistic competitions like “Rethink. Recreate. Reflect.” and guided young aspiring journalists with a better penmanship as Operations Content Head, and writer of Minerva Journalism Club, Jun'23 – Jan'24
- Published three articles in Centre for Innovation & Entrepreneurship, Online Newsletter Edition 6, and 7, Aug'22 – Jun'23
- Worked as a student tutor for Grade 8, Mathematics at Athena Mentorship, Jul'22 - Jul'23